CALCULUS · DERIVATIVES

Secant to Tangent

Explore & Reflect

Name _____ Date _____ Period _____

Open **Secant to Tangent** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

1 Predict before you touch the tool

1. For $f(x) = x^2$, what is the slope of the secant line from $x = 1$ to $x = 3$?

2. As the second point moves toward $x = 1$, what do you think happens to the secant slopes?

2 Explore use the tool

1. $f(x) = x^2$, $a = 1$. Record the secant slope for $h = 1$, 0.1 and 0.01.

2. Now use $h = -0.1$ and $h = -0.01$. What do the slopes approach from the left?

3. For $|x|$ at $a = 0$, record the slopes for $h = 0.01$ and $h = -0.01$. Why is there no derivative there?

4. For $\sin x$, visit at least 8 points and build the slope graph. Which function does it look like?

3 Record use the tool and fill in the table

Secant slopes from both sides

Round to 4 decimal places.

f(x)	a	slope with h = 0.01	slope with h = −0.01	f′(a)
x^2	2			
$x^3 - 3x$	1			
e^x	0			
$\sin x$	0			

4 Explain what did you discover?

1. The slope of the tangent line at a is the _____ of the secant slopes as h → _____.

2. h can't equal 0, because a secant line needs _____.

3. f is not differentiable at a corner, because the left and right secant slopes approach _____.

4. Where f has a horizontal tangent line, $f'(a)$ = _____.

5. Explain how the graph of f' for $f(x) = x^2$ is built from tangent slopes, and why it is a straight line.

5 Apply a new situation, no tool

1. Use the limit definition to find $f'(3)$ for $f(x) = x^2 - x$.

2. $f(1.9) = 6.62$, $f(2) = 7$, $f(2.1) = 7.42$. Estimate $f'(2)$.

3. Where is $g(x) = |x - 3|$ not differentiable? Why?

4. For $f(x) = x^3 - 3x$, where is the tangent line horizontal?